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# Patent ductus arteriosus (PDA) in preterm infants: Management and outcome

**AUTHOR:** Joseph B Philips III, MD, FAAP
**SECTION EDITORS:** Joseph A Garcia-Prats, MD, David R Fulton, MD
**DEPUTY EDITOR:** Niloufar Tehrani, MD

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## INTRODUCTION

During fetal development, the ductus arteriosus (DA) diverts blood from the pulmonary artery into the aorta, thereby bypassing the lungs ( figure 1). After birth, the DA undergoes active constriction and eventual obliteration. A patent ductus arteriosus (PDA) ( figure 2) occurs when the DA fails to completely close after delivery.

The management of PDA in preterm infants will be reviewed here. The clinical features and diagnosis of PDA in preterm infants are reviewed separately. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis".)

The diagnosis and management of PDA in term infants, children, and adults are discussed separately. (See "Clinical manifestations and diagnosis of patent ductus arteriosus (PDA) in term infants, children, and adults" and "Management of patent ductus arteriosus (PDA) in term infants, children, and adults".)

## DEFINITIONS

- **Definitions of prematurity** – Definitions of different degrees of prematurity based upon gestational age (GA; which is calculated from the first day of the mother's last period) or birth weight (BW) are provided in the table ( table 1).

- **Hemodynamically significant PDA** – The definition that we use to define a hemodynamically significant PDA is based upon both clinical findings and echocardiographic measurements. The assessment is based upon multiple findings in combination rather than a single isolated finding. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Hemodynamically significant PDA'.)

  - Clinical findings that indicate a hemodynamically significant PDA include need for substantial respiratory support, cardiomegaly and/or pulmonary edema on chest radiograph, hypotension requiring vasopressor support, and evidence of poor perfusion (lactic acidosis, oliguria) ( table 2).

  - Echocardiographic parameters include PDA diameter >1.5 mm, left atrial and/or ventricular enlargement, and diastolic flow reversal in the abdominal aorta ( table 3).

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## TREATMENT APPROACHES

Preterm infants with PDA should be managed collaboratively with input from the neonatal intensive care unit (NICU) team, pediatric cardiologist, and the parents/caregivers. Management decisions should be based on the neonate's entire clinical picture, and not solely on the presence and size of the PDA.

**Different approaches** — There is variability among centers regarding the management of PDAs in preterm infants and the optimal approach is uncertain [1-3]. The two main management approaches include:

- Expectant management with general supportive measures alone initially. (See 'Expectant management' below and 'General measures' below.)

- Early pharmacologic therapy using cyclooxygenase (COX) inhibitors (eg, ibuprofen, indomethacin) or acetaminophen (paracetamol). (See 'Pharmacologic therapy' below.)

Other interventions that play a more limited role include:

- Surgical ligation, which is generally reserved for infants with substantial respiratory morbidity who either fail or cannot receive pharmacologic therapy. (See 'Surgical ligation' below.)

- Transcatheter closure, which is performed at some specialized centers with expertise and experience performing interventional transcatheter techniques in preterm infants. (See

'Transcatheter occlusion' below.)

The strategy of prophylactic pharmacologic therapy given to all high-risk extremely preterm (EPT) infants even in the absence of echocardiographic evidence of a hemodynamically significant PDA is **not** recommended. Prophylactic therapy does not appear to reduce the morbidity associated with PDA, and it unnecessarily exposes many infants who would never have developed clinically significant PDAs to the potential harms associated with pharmacologic therapy. (See 'Prophylactic therapy' below.)

**Our approach** — At the author's center, we use a step-wise management approach for preterm infants with PDA that begins with supportive care alone in most cases (algorithm 1). We prefer this approach because spontaneous closure will occur in many patients. (See 'Expectant management' below and 'General measures' below.)

For infants with persistent hemodynamically significant PDAs who remain ventilator-dependent at one week of age, we proceed with a trial of pharmacologic therapy (algorithm 1). We typically use ibuprofen in this setting. Other centers use acetaminophen (paracetamol) or indomethacin. (See 'Pharmacologic therapy' below.)

If the infant fails two courses of pharmacologic therapy and remains ventilator-dependent, we proceed with a closure procedure (either transcatheter occlusion or surgical ligation) (algorithm 1). In our experience with our current management approach, this is an uncommon scenario. For neonates who require a closure procedure, transcatheter occlusion is the preferred procedure at our institution given the availability of experienced pediatric interventional cardiologists with expertise in this procedure in preterm neonates. In centers without such expertise, surgical ligation is preferred. Surgical ligation is used at our center for infants who are not candidates for transcatheter closure due to severity of lung disease or other practical limitations (eg, size or anatomic limitations). (See 'Procedures to close the PDA' below.)

**Comparison of approaches** — The available clinical trial data comparing expectant management versus early pharmacologic treatment in preterm infants with large PDA have demonstrated that while pharmacologic treatment results in earlier closure of the PDA, other clinical outcomes (mortality, bronchopulmonary dysplasia [BPD], necrotizing enterocolitis [NEC]) are generally similar between the two approaches [4-10].

In a meta-analysis of six trials including 500 EPT infants with hemodynamically significant PDA (variably defined; the two most common echocardiographic criteria were diameter >1.5 mm and/or left atrial enlargement), mortality rates were similar in those managed with expectant management compared with early (within seven days after birth) pharmacologic treatment (11 versus 9 percent; relative risk [RR] 1.25, 95% CI 0.72-2.17) [9]. Rates of BPD were also similar in

both groups (26 versus 24 percent; RR 1.11, 95% CI 0.78-1.6). Stage 2 or greater NEC occurred less commonly in the expectant management group (3 versus 7 percent), but the difference was not statistically significant (RR 0.43, 95% CI 0.16-1.2). In a separate analysis of seven trials (384 participants) comparing expectant management with very early (within 72 hours after birth) pharmacologic treatment, mortality rates were similar in both groups (17 versus 16 percent; RR 1.06, 95% CI 0.65-1.72) as were rates of BPD and other morbidities [9].

Additional support for the expectant management approach comes from five subsequent clinical trials not included in the meta-analyses [4,6-8,10]. All five trials allowed for rescue therapy, though the criteria for rescue therapy varied. Taken together, these data suggest that while early pharmacologic treatment results in earlier closure of the PDA, this does not appear to convey a meaningful clinical benefit with regard to survival, neonatal morbidities, or long-term neurodevelopmental outcomes.

The largest trial to address this question was the Baby-OSCAR trial, which involved 653 EPT infants with large PDA who were randomly assigned to early ibuprofen therapy or placebo starting within 72 hours after birth [10]. After the initial course of the study drug, infants in both arms could be treated with open-label medical or surgical treatment if they had symptoms attributable to PDA. At three weeks of age, more infants in the early ibuprofen group had a closed or small PDA (56 versus 37 percent; RR 1.50, 95% CI 1.26-1.79). However, the trial did not detect a difference in mortality at 36 weeks postmenstrual age (13.6 percent in the early treatment group versus 10.3 percent in the placebo group; RR 1.32, 95% CI 0.92-1.90) nor in rates of moderate-to-severe BPD (64 versus 59 percent; RR 1.09, 95% CI 0.96-1.23). Both groups also had similar rates of other neonatal morbidities, including grade III or IV intraventricular hemorrhage (IVH) (14 versus 11 percent), retinopathy of prematurity (ROP) (14 percent in both groups), severe NEC (13 percent in both groups), and pulmonary hemorrhage (8 versus 6 percent). Patients in the early treatment group less frequently received rescue medical therapy (13 versus 26 percent) or surgical intervention for PDA (3 versus 10 percent).

Similar findings were reported in four earlier randomized trials [4,6-8].

Long-term outcomes were reported in one trial involving 337 EPT infants with large PDA who were randomly assigned to early ibuprofen therapy or placebo starting within 12 hours after birth [7]. At 24 months corrected age, the number of patients who were alive without cerebral palsy was similar in both groups (71 percent in the early treatment group versus 72 percent in the placebo group; RR 0.98, 95% CI 0.83-1.16).

Observational studies comparing outcomes in neonates managed with expectant versus pharmacologic treatment have reached variable conclusions [11-23]. Differences in the findings

in these studies may be due to differences in the population studied (eg, gestational age, severity of illness), criteria used for intervention, and the era in which they were conducted.

**Goals of treatment** — The goals of management are to:

- Reduce morbidity from the effects of excessive left-to-right shunting and ductal steal. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Consequences of a PDA'.)
- Avoid unnecessary treatment in neonates who will likely have spontaneous closure without any adverse sequelae of the PDA.
- Ensure timely treatment of PDA in infants for whom the PDA is likely contributing to significant respiratory disease and/or other neonatal morbidity.

Identifying which infants are most likely to benefit from intervention remains a challenge. (See 'Unanswered questions' below.)

In a multicenter international study involving nearly 40,000 EPT neonates managed at 139 NICUs, there was U-shaped relationship between rates of treatment for PDA (pharmacologic or surgical) at each NICU and mortality rates, with higher mortality at centers with the lowest and highest rates of treatment for PDA [24]. A similar U-shaped relationship was seen for neurologic outcome. These U-shaped relationships suggest that if the threshold for intervention is too high, there can be adverse consequences, likely due to the hemodynamic effects of untreated significant PDA. At the same time, if the threshold for intervention is too low, there may be adverse consequences related to unnecessary treatment of infants with mild or clinically insignificant PDA.

**Unanswered questions** — Ongoing challenges and unanswered questions in the management of PDA in preterm neonates include:

- What criteria should be used to define hemodynamically significant PDA? The definition is not standardized and there was considerable variability in the definitions used in the clinical trials discussed above. We generally favor a definition that includes both clinical and echocardiographic criteria. This is discussed in greater detail separately. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Hemodynamically significant PDA'.)
- Is it possible to identify an at-risk population who may benefit from early pharmacologic treatment? For example, some have advocated for early treatment in the most premature neonates (eg, GA <25 weeks) since these neonates are at high risk of experiencing

complications, and the likelihood of spontaneous closure is lower than in more mature neonates. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Spontaneous ductal closure after birth'.)

- If an expectant management approach is used, what threshold should be used for intervention if the PDA does not close spontaneously? The clinical trials described above used variable criteria for providing rescue therapy. As a result, use of rescue therapy varied widely in these trials, from as low as 6 percent to as high as 62 percent [4,6-8]. The optimal threshold remains uncertain.

- When pharmacologic treatment is used, what endpoint should be targeted (is complete ductal closure necessary or is improvement in clinical status sufficient)?

- What criteria should be used for performing a closure procedure (transcatheter occlusion or ligation)?

## GENERAL MEASURES

General measures that are provided in all neonates with PDA include:

- **Neutral thermal environment** – A neutral thermal environment that minimizes metabolic demands on ventricular function. (See "Overview of short-term complications in preterm infants", section on 'Hypothermia'.)

- **Respiratory support** – Respiratory support is tailored according to the neonate's degree of respiratory compromise. Details are provided separately. (See "Respiratory support, oxygen delivery, and oxygen monitoring in the newborn" and "Respiratory distress syndrome (RDS) in preterm infants: Management", section on 'Ongoing respiratory support' and "Bronchopulmonary dysplasia (BPD): Management and outcome", section on 'Respiratory support'.)

  For infants who require mechanical ventilation, care should be taken to minimize the risk of ventilator-induced lung injury, as summarized in the table ( table 4) and discussed in detail separately. (See "Approach to mechanical ventilation in very preterm neonates".)

- **Monitoring for signs of anemia** – The thresholds for transfusion in infants with PDA are the same as for the general preterm neonatal population, as summarized in the figure ( table 5) and discussed in detail separately. (See "Red blood cell (RBC) transfusions in the neonate".)

- **Fluid restriction with or without diuretic therapy** – Excessive fluid administration (>170 mL/kg per day) is associated with an increased incidence of PDA [25]. Our usual practice is to restrict total fluid intake to 130 to 140 mL/kg per day. A more restrictive limit may be warranted if there are overt signs of pulmonary edema. The fluid status of the patient must be monitored frequently to avoid under- or overhydration as fluid needs widely vary in preterm infants due to differences in insensible fluid loss. (See "Fluid and electrolyte therapy in newborns" and "Respiratory distress syndrome (RDS) in preterm infants: Management", section on 'Fluid management'.)

  Use of diuretic therapy is generally limited to infants with significant respiratory morbidity (ie, bronchopulmonary dysplasia [BPD]), as discussed separately. (See "Bronchopulmonary dysplasia (BPD): Management and outcome", section on 'Diuretics'.)

  If diuretic therapy is used, furosemide and other loop diuretics are generally avoided in the first one to two weeks after birth. This is because loop diuretics stimulate renal synthesis of prostaglandin $E_2$, a potent vasodilator that maintains ductal patency [26].

## EXPECTANT MANAGEMENT

Expectant management consists of the supportive care measures described above and close clinical monitoring. (See 'General measures' above.)

Most neonates managed expectantly will have spontaneous closure of the PDA. The expected time to spontaneous closure increases with decreasing gestational age [21]. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Spontaneous ductal closure after birth'.)

However, some infants may require rescue therapy due to ongoing moderate to severe symptoms. The optimal threshold for starting therapy in this setting is uncertain and practice varies. Our suggested approach is as follows:

- **Clinically stable neonates** – Neonates who remain clinically stable with only modest respiratory support requirements (ie, continuous positive airway pressure, other noninvasive ventilation modality, or less intensive support) and without signs of systemic hypoperfusion (eg, oliguria, acidosis) generally do not require any intervention to close the PDA. Follow-up echocardiogram is performed at 36 weeks postmenstrual age (or sooner if any concerns arise). If the PDA has closed by that time, no further evaluation is necessary. If the PDA persists, a repeat echocardiogram is performed before neonatal intensive care unit (NICU) discharge to determine need for post-discharge follow-up.

- **Neonates with clinical concern for persistent hemodynamically significant PDA** – Neonates who remain ventilator-dependent at one week of age and those with hemodynamic instability attributable to the PDA should have a repeat echocardiogram performed to confirm that there is a persistent hemodynamically significant PDA. If the echocardiogram confirms persistence of a hemodynamically significant PDA, we proceed with a trial of pharmacologic therapy (see 'Pharmacologic therapy' below). If the echocardiogram findings suggest only a small or trivial PDA, the neonate should be evaluated for other causes of their clinical instability.

The evidence supporting expectant management is summarized above. (See 'Comparison of approaches' above.)

Our center uses a seven-day threshold for initiating medical therapy in ventilator-dependent neonates with PDA. Other centers may use different thresholds (eg, some may start treatment at 72 hours in intubated high-risk extremely preterm neonates). The rationale for the threshold we use at our center is based upon observational data suggesting that the risk of bronchopulmonary dysplasia (BPD) in mechanically ventilated neonates increases sharply when the PDA persists for ≥7 days [27]. These data are discussed separately. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Consequences of a PDA'.)

## PHARMACOLOGIC THERAPY

Pharmacologic therapy is directed towards inhibiting prostaglandin synthetase, as prostaglandin $E_2$ ($PGE_2$), which has two catalytic sites, promotes ductal patency. The nonselective cyclooxygenase (COX) inhibitors (ibuprofen and indomethacin) inhibit the COX site, whereas acetaminophen (paracetamol) affects the peroxidase segment.

**Indications** — As previously discussed, different neonatal intensive care units (NICUs) use different thresholds for initiating pharmacologic therapy for treatment of PDA in preterm neonates. In many NICUs (including the author's institution), initial management is supportive, and pharmacologic therapy is reserved for symptomatic infants with persistent hemodynamically significant PDAs ( algorithm 1). However, other NICUs prefer a strategy of early treatment in which pharmacologic therapy is used in any extremely preterm (EPT) neonate diagnosed with a large PDA in the first 24 to 48 hours after birth [28]. The optimal approach is uncertain. (See 'Treatment approaches' above and 'Comparison of approaches' above.)

**Choice of regimen** — The choice of regimen is highly center-specific [3]. All three agents appear to have similar efficacy, though ibuprofen and acetaminophen likely have fewer adverse effects compared with indomethacin [29-31]. (See 'Comparative data' below.)

Considerations that factor into decision-making include:

- Cost and availability of each drug
- Whether the neonate is tolerating enteral feeds
- Whether the neonate has any absolute or relative contraindications to treatment with a nonselective COX inhibitor (active bleeding, acute kidney injury [AKI]) (see 'Contraindications' below)

At the author's center, when pharmacologic therapy is used, we preferentially use standard-dose ibuprofen (10 mg/kg for the first dose administered intravenously [IV] or enterally, followed by two doses of 5 mg/kg given every 24 hours). A second course is given if the follow-up echocardiogram shows failure of PDA closure and the infant is still ventilator-dependent ( algorithm 1). (See 'Ibuprofen' below and 'Response to therapy' below.)

For patients with contraindications to ibuprofen, we use acetaminophen (paracetamol). (See 'Contraindications' below and 'Acetaminophen (paracetamol)' below.)

Acetaminophen is increasingly used as a first-line agent at many centers. By contrast, use of indomethacin has declined since the early 2000s given the availability of IV ibuprofen and acetaminophen, which have fewer adverse effects. (See 'Acetaminophen (paracetamol)' below and 'Indomethacin' below.)

**Comparative data**

- **Ibuprofen versus indomethacin** – Ibuprofen and indomethacin appear to have similar efficacy for achieving PDA closure, but the risk of adverse effects (acute kidney injury [AKI], necrotizing enterocolitis [NEC]) is lower with ibuprofen [1,29]. In a meta-analysis of 24 clinical trials (1590 neonates), ibuprofen (administered IV or enterally) and indomethacin had similar efficacy for achieving PDA closure after the first treatment course (success rates of 70 and 72 percent, respectively; relative risk [RR] 0.93, 95% CI 0.81-1.1) [29]. Infants treated with ibuprofen were less likely to develop NEC (7 versus 11 percent; RR 0.68, 95% CI 0.49-0.94) or oliguria (3 versus 12 percent; RR 0.28, 95% CI 0.14-0.54). Mortality was similar in both groups.

- **Ibuprofen versus acetaminophen (paracetamol)** – Ibuprofen and acetaminophen appear to have similar efficacy for achieving PDA closure; however, the risk of occult

gastrointestinal bleeding appears to be higher with ibuprofen [31-36]. In a meta-analysis of five trials (559 neonates), ibuprofen and acetaminophen had similar efficacy for achieving PDA closure after the first treatment course (success rates of 67 and 69 percent, respectively; RR 0.95, 95% CI 0.75-1.2) [31]. Infants treated with ibuprofen had higher creatinine levels and were more likely to develop occult blood in their stools (8 versus 2 percent; RR 3.6, 95% CI 1.4-8.3). The clinical significance of these findings is debatable. Mortality was similar in both groups.

- **Acetaminophen (paracetamol) versus indomethacin** – Based on limited clinical trial data, acetaminophen and indomethacin appear to have similar efficacy for achieving PDA closure [31]; the risk of adverse effects is likely higher with indomethacin. In a meta-analysis of two trials (273 neonates), acetaminophen and indomethacin had similar efficacy for achieving PDA closure by the end of treatment (success rates of 85 percent in both groups) [31].

The relative efficacy of different regimens used to treat PDA in preterm neonates has been examined in network meta-analyses, which estimate treatment effects based upon direct and indirect comparisons in the available clinical trials [37,38]. In a network meta-analysis of 68 randomized trials comparing various treatment regimens (including indomethacin, standard- and high-dose IV ibuprofen, standard- and high-dose oral ibuprofen, oral acetaminophen [paracetamol], and placebo [ie, expectant management]), high-dose oral or IV ibuprofen appeared to be among the most effective regimens for achieving PDA closure within seven days of starting therapy (based upon estimates from 17 direct comparisons and 45 indirect comparisons) [37]. For example, high-dose oral ibuprofen was more likely to achieve PDA closure compared with IV indomethacin (odds ratio [OR] 2.35, 95% CI 1.08-5.31). The differences were not statistically significant when compared with oral acetaminophen (OR 1.23, 95% CI 0.62-2.48) or standard-dose oral ibuprofen (OR 1.63, 95% CI 0.84-3.24). There were no statistically significant differences in mortality or rates of bronchopulmonary dysplasia (BPD) or NEC between any of the different treatment regimens, including placebo (expectant management). An important limitation of these data is that the analyses rely on the assumption of transitivity (ie, that the trial populations, interventions, cointerventions, and assessment of outcomes were sufficiently similar in all of these trials). Given the large number of included clinical trials that spanned multiple different eras, the transitivity assumption is unlikely to hold up. Thus, the certainty of these findings is low.

### Specific agents

#### Ibuprofen

**Dose** — Ibuprofen can be administered enterally or IV using a standard- or high-dose regimen. In our practice, we use the standard-dose regimen. The available clinical trial data suggest that the high-dose regimen is more effective for achieving PDA closure at the end of treatment; however, it is also associated with higher risk of adverse effects (eg, AKI, gastrointestinal bleeding, NEC) [29]. (See 'Comparative data' above.)

- Standard-dose regimen – The standard-dose regimen consists of an initial dose of 10 mg/kg given enterally or IV followed by two additional doses of 5 mg/kg given at 24-hour intervals.

- High-dose regimen – The high-dose regimen consists of an initial dose of 15 to 20 mg/kg given enterally of IV followed by two additional doses of 7.5 to 10 mg/kg administered at 12- to 24-hour intervals.

IV and enteral administration of ibuprofen appear to be equally effective for PDA closure [29]. In resource abundant settings, the IV product is typically used for treating PDA in preterm neonates. However, IV ibuprofen is costly and may not be available in all settings. Thus, some neonatal units may prefer the enteral product.

**Contraindications** — The following contraindications apply to both ibuprofen and indomethacin:

- Absolute contraindications:

  - AKI (serum creatinine >1 mg/dL [88.4 micromol/L])
  - Active bleeding, especially active intracranial hemorrhage or gastrointestinal bleeding
  - NEC (proven or clinically suspected)
  - Ductal-dependent congenital heart disease (CHD)

- Relative contraindications – We also generally avoid ibuprofen (and indomethacin) in the following scenarios:

  - Thrombocytopenia and/or coagulopathy – We do not routinely measure platelet counts or coagulation studies prior to drug administration if there is no clinical concern for coagulopathy. However, if the infant has documented thrombocytopenia or coagulopathy, we avoid ibuprofen (and indomethacin).

  - Acute infection (proven or suspected).

  - Oliguria (urine output <1 cc/kg per hour).

In most of these circumstances, if the infant requires pharmacotherapy for PDA closure, we use acetaminophen (paracetamol) (see 'Acetaminophen (paracetamol)' below). The exception is infants with ductal-dependent CHD since they generally require medical therapy to maintain ductal patency rather than therapy to close the PDA. In fact, closure of the PDA in a newborn with ductal-dependent CHD can be life-threatening. (See "Identifying newborns with critical congenital heart disease", section on 'Ductal-dependent lesions'.)

**Adverse effects** — Potential adverse effects associated with nonselective COX inhibitors include [39-41]:

- AKI
- Bleeding, particularly gastrointestinal bleeding
- NEC
- Spontaneous intestinal perforation
- Hyperbilirubinemia, due to displacement of bilirubin from binding sites on albumin; however, this has been reported only with drug levels that far exceed those used clinically [42-45]

Ibuprofen is associated with a lower risk of gastrointestinal and renal side effects than indomethacin [29]; however, these remain potential risks associated with both drugs. Some of these adverse effects occur due to reduced cerebral, gastrointestinal, and renal blood flow caused by nonselective COX inhibitors. This blood flow reduction is greater with indomethacin compared with ibuprofen, which may explain why ibuprofen has a better safety profile [46-49].

**Feeding during treatment** — Because of the potential adverse effects of ibuprofen or indomethacin on the gastrointestinal tract, there have been concerns about initiation or continuation of enteral feeds during administration of either drug. However, several studies have shown that enteral feeding during treatment is safe [50-52]. As a result, we continue enteral feedings while on treatment.

**Efficacy** — The efficacy of ibuprofen for treatment of PDA in preterm neonates is supported by clinical trials and meta-analyses [29,53]. These data are discussed above:

- Compared with placebo (expectant management) (see 'Comparison of approaches' above)
- Compared with other agents (see 'Comparative data' above)

**Indomethacin** — Indomethacin is usually given IV and more than one dose is typically required for sustained constriction. Various dosing regimens have been described, ranging from 0.1 and 0.2 mg/kg per dose administered at 12- to 24-hour intervals.

Commonly used dosing regimens include [37,54,55]:

- Standard weight-based dosing – Three doses of 0.2 mg/kg per dose given IV at 12-hour intervals.
- Dosing based on postnatal age:
  - Infants <48 hours of age – The first dose is 0.2 mg/kg IV for the first dose, followed by two additional doses of 0.1 mg/kg given IV at 12-hour intervals.
  - Infants >48 hours and <7 days – Three total doses of 0.2 mg/kg per dose given IV at 12-hour intervals.
  - Infants ≥7 days of age – Three total doses of 0.25 mg/kg per dose given IV at 12-hour intervals.

More prolonged treatment courses (ie, >3 doses) should generally be avoided since this increases the risk of NEC and does not appear to increase the likelihood of PDA closure [56].

Measuring plasma drug levels is not necessary and this testing is not widely available.

Contraindications to indomethacin are the same as for ibuprofen. The approach to enteral feeding while on treatment is also the same. (See 'Contraindications' above and 'Feeding during treatment' above.)

Adverse effects are generally similar to those of ibuprofen. However, indomethacin is associated with a higher risk of AKI and NEC compared with ibuprofen [1,29]. (See 'Adverse effects' above.)

**Acetaminophen (paracetamol)** — Accumulating evidence suggests that both oral and IV acetaminophen are reasonable alternatives to nonselective COX inhibitor therapy in preterm infants with PDA [31,57]. In our practice, we use acetaminophen for patients in whom nonselective COX inhibitor therapy is contraindicated (see 'Contraindications' above). Other centers use acetaminophen as the first-line agent. (See 'Choice of regimen' above and 'Comparative data' above.)

The dosing regimen for PDA closure consists of 15 mg/kg per dose given IV or enterally every six hours for three to seven days [31,34,58].

After the third day of therapy, an echocardiogram is performed to see if there has been a response to therapy. If the PDA is closed, therapy can be discontinued. However, if it remains open, acetaminophen is continued for a full seven days of treatment. Infants requiring a seven-day course should have liver function tests monitored.

**Response to therapy** — The response to therapy is assessed based on the neonate's clinical status and repeat echocardiogram performed 24 to 48 hours after completion of the treatment course. For neonates treated with acetaminophen (paracetamol), an echocardiogram is obtained after the third day of treatment to determine if a full seven-day treatment course is necessary. If the PDA has not closed, treatment is continued, and a follow-up echocardiogram is obtained after treatment is completed.

Once treatment is completed, if the PDA has closed or reduced to small or trivial size, treatment is considered successful, and no further intervention is required.

If the infant continues to have a hemodynamically significant PDA and remains symptomatic (eg, ongoing ventilator dependence), treatment is considered unsuccessful and a second treatment course is administered, as discussed below. (See 'Treatment failure' below.)

**Treatment failure** — Approximately 25 to 40 percent of medically treated patients fail to respond to the initial treatment course [29-31].

- **Risk factors** – Risk factors for treatment failure include [59,60]:
  - Lower gestational age (GA)
  - Lack of exposure to antenatal glucocorticoid therapy
  - Severe respiratory disease
  - Postnatal age >1 month
- **Retreatment** – If the infant continues to have a hemodynamically significant PDA and remains symptomatic (eg, ongoing ventilator dependence), a second course of treatment can be given either using the same agent or a different agent. Approximately 40 to 50 percent of patients who fail initial therapy will respond to a second course [61-63].

  Studies on the use of combination therapy (eg, ibuprofen plus acetaminophen) for patients with persistent PDA have reached variable conclusions [64,65]. Taken together, the available studies suggest that combination therapy probably does not result in higher rates of PDA closure compared with single-agent therapy [64]. Thus, we suggest not using this approach.
- **Refractory patients** – For patients who fail to respond to two courses of therapy, subsequent treatment courses are unlikely to close the PDA. If the infant remains symptomatic from the PDA (eg, ventilator-dependent), a definitive closure procedure (surgical ligation or transcatheter occlusion) is generally warranted. (See 'Procedures to close the PDA' below.)

## PROCEDURES TO CLOSE THE PDA

Procedural closure options include transcatheter occlusion and surgical ligation. In the modern era, these procedures are seldom required and are reserved for infants with symptomatic large PDAs who fail to respond to pharmacologic therapy.

**Surgical ligation** — Surgical ligation is performed via posterolateral thoracotomy. Some surgeons perform multiple ligations or divisions to minimize the risk of recurrent shunt [66].

Postoperative care includes continuous cardiovascular monitoring and hemodynamic support to maintain adequate blood pressure and perfusion [67]. Serial echocardiography may be used to assess postoperative cardiac function and guide management.

Risks associated with PDA ligation include [11,68,69]:

- Hemodynamic instability and low cardiac output (sometimes called post-ligation cardiac syndrome) [70]
- Hemorrhage
- Respiratory deterioration
- Infection
- Intraventricular hemorrhage (IVH)
- Chylothorax
- Recurrent laryngeal nerve paralysis
- Pneumothorax

It remains uncertain whether the surgical ligation procedure itself is a major contributor to morbidity and mortality. Most patients who undergo surgical ligation are more severely compromised to begin with and thus outcomes for this population are generally less favorable compared with those who do not require PDA ligation. (See 'Outcome' below.)

In a multicenter retrospective study of 754 infants <28 weeks GA with hemodynamically significant PDA managed medically (n = 570) or with surgical ligation (n = 184), those who underwent ligation had higher rates of neonatal comorbidities prior to treatment (eg, mechanical ventilation dependence, sepsis, necrotizing enterocolitis) [71]. Fewer patient in the surgical ligation survived to age 18 to 24 months without moderate to severe neurologic disability (40 versus 48 percent); however, after adjusting for other confounders, the difference was not statistically significant (adjusted odds ratio (aOR) 0.97; 95% CI 0.65-1.44).

Similarly, a retrospective study from a single center of 166 extremely preterm (EPT) infants who underwent surgical ligation and 142 EPT infants who did not found that, after adjusting for other confounders, PDA ligation was not independently associated with increased risk of adverse outcome (neurodevelopmental impairment, bronchopulmonary dysplasia [BPD], or retinopathy of prematurity [ROP]) [72]. In a follow-up report from the same center, 157 of 166 EPT infants treated with surgical ligation survived and were successfully extubated (median 12 days after ligation) [73]. Of note, earlier extubation was associated with larger PDA diameter and moderate to severe left ventricular dilatation, suggesting that the PDA was a significant contributor to the respiratory failure in these infants.

**Transcatheter occlusion** — Transcatheter PDA occlusion has been performed in both term and preterm infants, including preterm extremely low birth weight (ELBW; <1000 g) neonates [74-79]. The procedure is usually performed using access through the femoral artery or vein.

In 2019, the US Food and Drug Administration (FDA) approved the Amplatzer Piccolo Occluder for use in infants weighing >700 g and age >3 days [80].

Experience with this procedure is growing, and it is performed in some specialized centers with experienced pediatric interventional cardiologists [81].

In a systematic review and meta-analysis that included of 28 studies (mostly small retrospective case series) including a total of 373 infants ≤1.5 kg who underwent transcatheter PDA closure, the procedure was successfully completed in 96 percent of cases [82]. The pooled rate of major adverse events was 8 percent, including 5 deaths (1.3 percent) related to the procedure. Minor adverse events occurred in 18 percent.

The relative frequencies of procedure-related complications reported in studies of transcatheter device PDA closure with the Piccolo Occluder in ELBW infants include [83]:

- Device embolization (3 percent)
- Aortic obstruction or narrowing (1 percent)
- Obstruction of the left pulmonary artery (1 percent)
- Tricuspid regurgitation (3 percent)
- Vascular injury (1 percent)

Cardiorespiratory instability can occur after transcutaneous PDA closure, similar to the post-ligation cardiac syndrome described after surgical ligation [84].

## PROPHYLACTIC THERAPY

We suggest **not** routinely using prophylactic pharmacologic therapy to reduce the incidence of PDA. Prophylactic treatment does not appear to reduce the morbidity associated with PDA and it unnecessarily exposes infants without significant PDA to the potential adverse effects of pharmacologic therapy [9,53,85-90].

In a meta-analysis of eight trials (1791 extremely preterm [EPT] neonates), prophylactic indomethacin therapy reduced the need for invasive PDA closure (relative risk [RR] 0.51, 95% CI 0.37-0.71), but there was no impact on mortality or neurodevelopmental impairment (RR 1.02, 95% CI 0.90-1.15) [53]. Similarly, in a meta-analysis of seven trials (925 EPT neonates), prophylactic ibuprofen therapy reduced the need for invasive PDA closure (RR 0.46, 95% CI 0.22-0.96), without any clear benefit with regards to survival or neurodevelopmental outcome [53]. There are limited data on prophylactic acetaminophen therapy [91], but it is likely to have the same results as the other two agents.

## OUTCOME

Preterm infants with clinically significant PDAs have a higher risk of mortality and neonatal morbidity (eg, bronchopulmonary dysplasia [BPD], intraventricular hemorrhage [IVH], necrotizing enterocolitis [NEC]) compared with neonates of similar gestational age and birth weight without PDAs [92-94]. (See "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Consequences of a PDA'.)

As discussed above, the risk of mortality and morbidity appear to be similar regardless of whether the infant is managed expectantly or with early pharmacologic therapy. (See 'Comparison of approaches' above.)

Outcomes are generally worse for infants who require surgical ligation, which likely reflects that this is a more severely compromised patient population [11,13,95,96].

## SUMMARY AND RECOMMENDATIONS

- **Initial management approach** – For most preterm neonates with PDA, we suggest expectant supportive care initially rather than early pharmacologic therapy (**Grade 2C**). This is because spontaneous closure will occur in many cases. While pharmacologic treatment results in earlier closure of the PDA, this does not appear to convey a meaningful clinical benefit in terms of survival, neonatal morbidities, or long-term neurodevelopmental outcomes. However, the optimal management approach is uncertain,

and some centers prefer a strategy of early pharmacologic therapy. (See 'Expectant management' above and 'Comparison of approaches' above.)

- **Supportive care** – General supportive care measures provided to all preterm infants with PDA include a neutral thermal environment, optimal respiratory support, monitoring for anemia, and modest fluid restriction. (See 'General measures' above.)

- **Ongoing clinical concern for hemodynamically significant PDA** – For patients with a persistent hemodynamically significant PDA ( table 3) who remain ventilator-dependent at one week of age or with signs of hemodynamic instability (eg, hypotension requiring vasoactive therapy, oliguria, lactic acidosis) not otherwise explained, we suggest pharmacologic treatment to close the PDA rather than ongoing supportive care alone (**Grade 2B**). (See 'Indications' above.)

  - **Choice of agent** – We suggest either ibuprofen or acetaminophen (paracetamol) rather than indomethacin (**Grade 2C**). However, indomethacin is a reasonable option, particularly if the costs of intravenous (IV) ibuprofen and acetaminophen are a limitation. All three agents appear to have similar efficacy; indomethacin has a higher risk of adverse effects (eg, acute kidney injury [AKI], gastrointestinal bleeding, necrotizing enterocolitis [NEC]). (See 'Choice of regimen' above and 'Comparative data' above and 'Adverse effects' above.)

    Contraindications to ibuprofen and indomethacin include AKI/oliguria, gastrointestinal bleeding, NEC (proven or suspected), thrombocytopenia and/or coagulopathy, acute infection, and ductal-dependent congenital heart disease (CHD). In most of these circumstances, if the infant requires pharmacotherapy for PDA closure, acetaminophen (paracetamol) can be used. The exception is infants with ductal-dependent CHD who should not be given medications to close the PDA. (See 'Contraindications' above and "Identifying newborns with critical congenital heart disease", section on 'Ductal-dependent lesions'.)

  - **Dosing**

    - **Ibuprofen** – The standard-dose ibuprofen regimen consists of an initial dose of 10 mg/kg given enterally or IV followed by two additional doses of 5 mg/kg given at 24-hour intervals. The high-dose regimen consists of an initial dose of 15 to 20 mg/kg given enterally of IV followed by two additional doses of 7.5 to 10 mg/kg administered at 12- to 24-hour intervals. The high-dose regimen has a higher success rate but also has a higher risk of side effects. In our practice, we use the standard-dose regimen. (See 'Dose' above.)

- **Acetaminophen (paracetamol)** – 15 mg/kg per dose given IV or enterally every six hours for three to seven days. (See 'Acetaminophen (paracetamol)' above.)

- **Indomethacin** – Three doses of 0.2 mg/kg per dose given IV at 12-hour intervals. (See 'Indomethacin' above.)

- **Response to therapy** – The response to therapy is assessed based on the neonate's clinical status and repeat echocardiogram performed 24 to 48 hours after completion of the treatment course ( algorithm 1). For infants who continue to have a hemodynamically significant PDA and who remain symptomatic (eg, ongoing ventilator dependence), we suggest a second course of treatment rather than proceeding with an invasive closure procedure (**Grade 2C**). (See 'Response to therapy' above.)

- **Patients who fail medical therapy** – For patients who fail to respond to two courses of therapy, subsequent treatment courses are unlikely to close the PDA. If the infant remains symptomatic from the PDA (eg, ventilator-dependent), options for definitive closure include surgical ligation or transcatheter occlusion. The choice between these is dependent on the expertise available at the center. (See 'Procedures to close the PDA' above.)

- **Outcome** – Preterm infants with clinically significant PDAs have a higher risk of mortality and morbidity compared with neonates of similar gestational age and birth weight without PDAs. Outcomes are generally worse for infants who require surgical ligation, which likely reflects that this is a more severely compromised patient population. (See 'Outcome' above and "Patent ductus arteriosus (PDA) in preterm infants: Clinical features and diagnosis", section on 'Consequences of a PDA'.)

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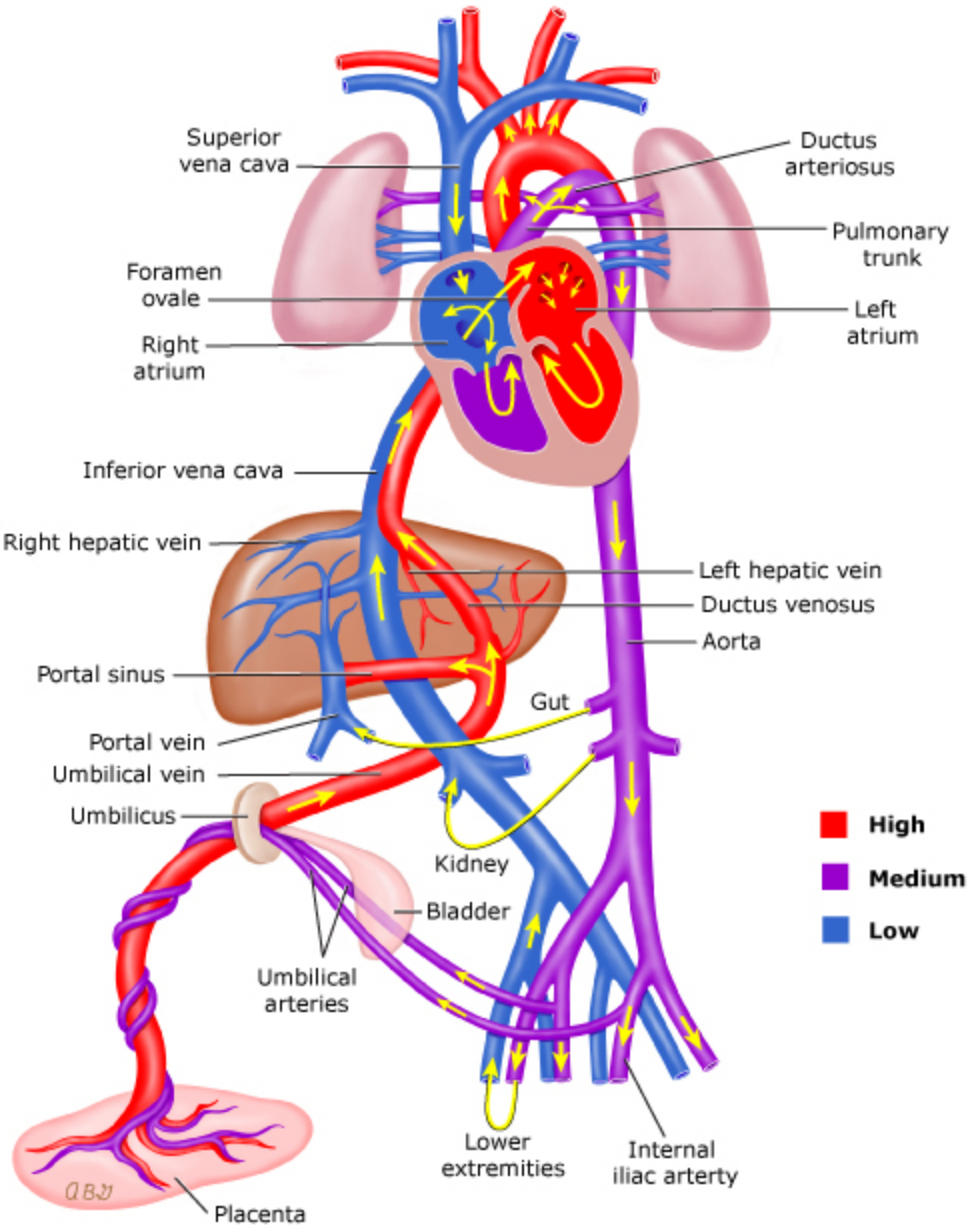
Topic 15814 Version 75.0

## GRAPHICS

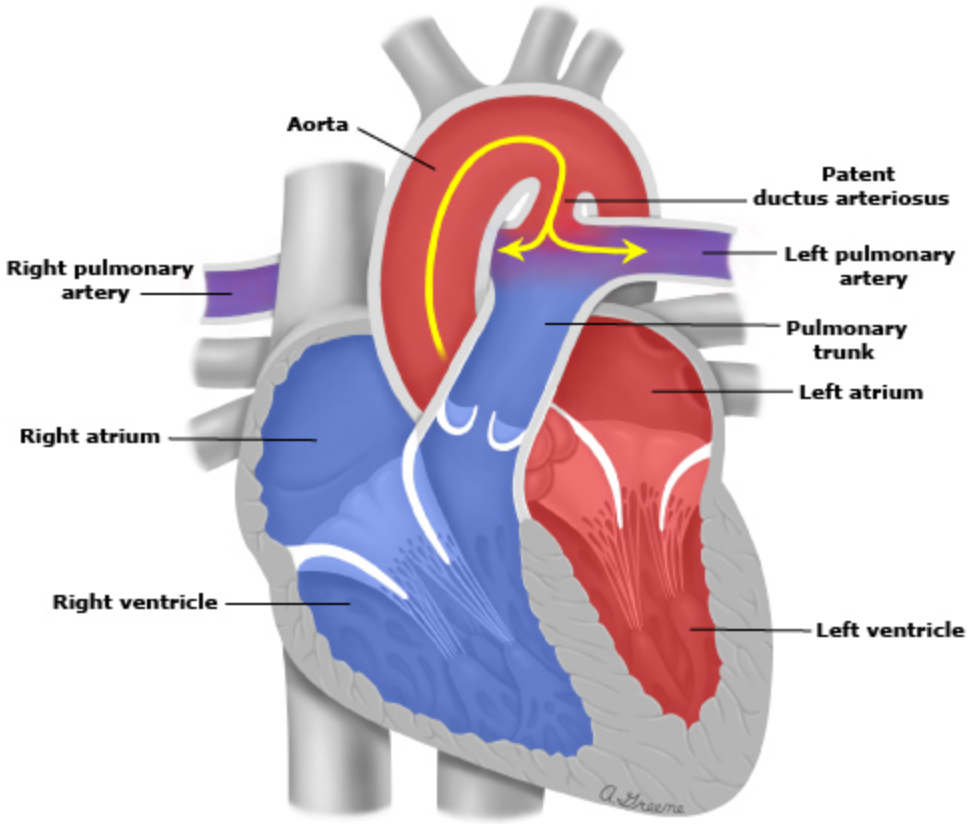
### Fetal circulation




The degree of oxygen saturation is indicated by shading, as explained in the figure key.

Graphic 66765 Version 4.0

## Patent ductus arteriosus




Graphic 65828 Version 2.0

## Classification of prematurity categorized by birth weight or gestational age

| | Birth weight |
|---|---|
| **Low birth weight (LBW)** | <2500 g |
| **Very low birth weight (VLBW)** | <1500 g |
| **Extremely low birth weight (ELBW)** | <1000 g |
| | **Gestational age** |
| **Term** | ≥37 weeks |
| **Late preterm** | 34 weeks to <37 weeks |
| **Moderate preterm** | 32 weeks to <34 weeks |
| **Very preterm** | <32 weeks |
| **Extremely preterm** | <28 weeks |

In using these definitions, the definition of VLBW infants includes ELBW infants, and the category of very preterm infants also includes those who are extremely preterm. This is an important consideration when one is reviewing published data of VLBW and very preterm infants.

Graphic 119362 Version 3.0

## Clinical findings that suggest a hemodynamically significant patent ductus arteriosus (PDA) in preterm infants

| Clinical findings |
| --- |
| Need for substantial respiratory support:<br>▪ MAP ≥8 cm $H_2O$, and/or<br>▪ $FiO_2$ >0.4 |
| Pulmonary hemorrhage |
| Widened pulse pressure (eg, DBP <50% of SBP or >25 mmHg difference between SBP and DBP) |
| Cardiomegaly and/or pulmonary edema on chest radiograph |
| Hypotension requiring vasoactive infusion(s) |
| Signs of hypoperfusion:<br>▪ Oliguria (urine output <1 mL/kg/hr) or AKI (Cr >1.2 mg/dL)<br>▪ Metabolic acidosis not otherwise explained |

This table summarizes clinical findings that suggest a hemodynamically significant PDA in preterm neonates. These findings should be assessed in conjunction with echocardiographic findings. The determination of whether a PDA is hemodynamically significant is typically based upon multiple findings rather than a single isolated finding. Refer to UpToDate topics on PDA in preterm infants for additional details.

PDA: patent ductus arteriosus; MAP: mean airway pressure; $FiO_2$: fraction of inspired oxygen; DBP: diastolic blood pressure; SBP: systolic blood pressure; AKI: acute kidney injury; Cr: creatinine.

Graphic 141521 Version 1.0

## Echocardiographic findings of hemodynamically significant patent ductus arteriosus (PDA) in preterm neonates

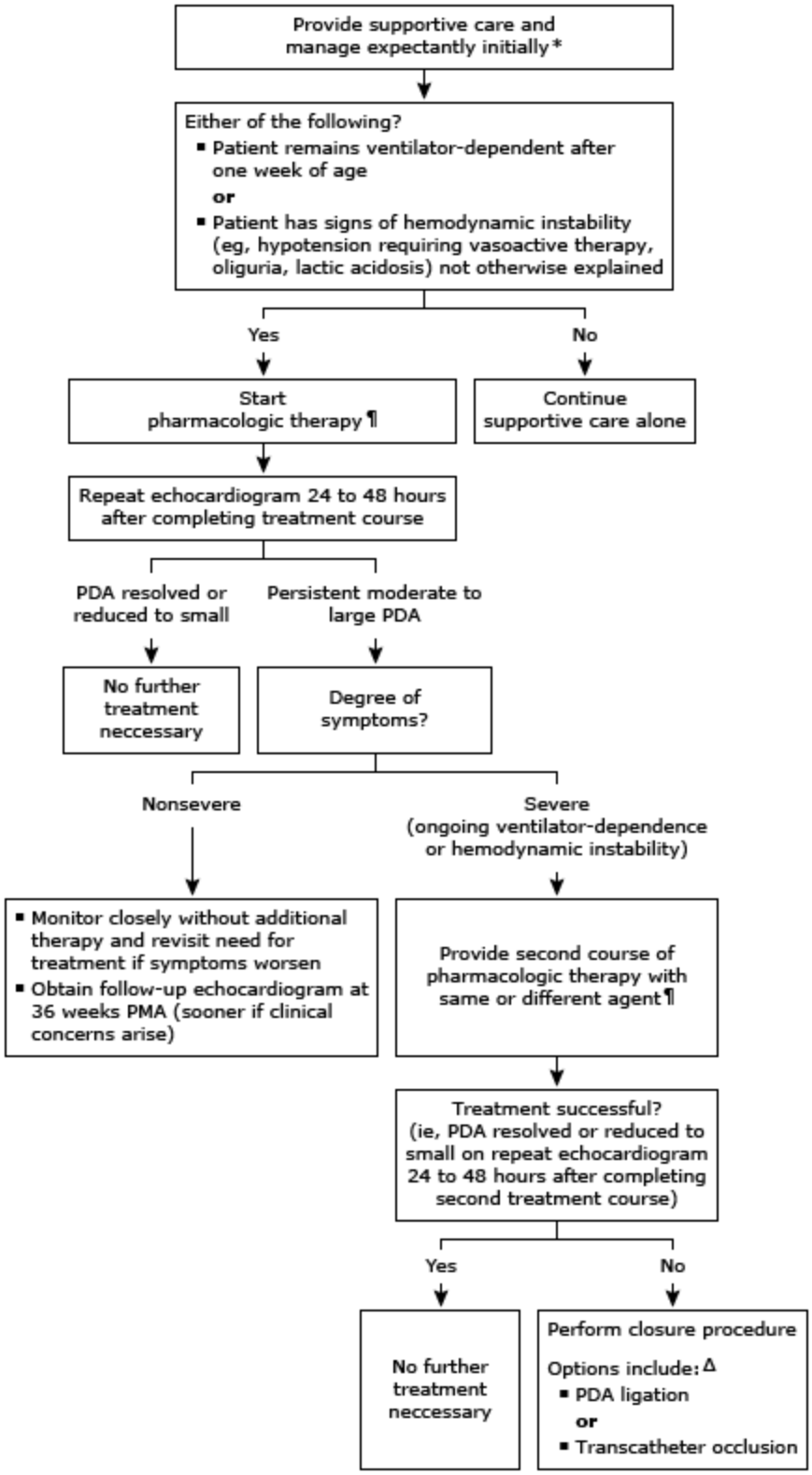
| Echocardiographic finding |
| --- |
| PDA diameter:<br>▪ Moderate size: 1 to 1.5 mm × √BSA **or** PDA:LPA ratio ≥0.5 to 1<br>▪ Large: >1.5 mm × √BSA **or** PDA:LPA ratio ≥1 |
| Diastolic flow reversal in the abdominal aorta |
| Unrestrictive flow across the PDA:<br>▪ Ao-PDA flow velocity <2 m/s, or<br>▪ Mean pressure gradient ≤8 mmHg across the PDA |
| LA enlargement not otherwise explained |
| LV enlargement not otherwise explained |
| LA hypertension (estimated qualitatively by septal bowing from LA to RA or quantitatively by flow velocity across PFO) |

This table summarizes echocardiographic findings that suggest a large and hemodynamically significant PDA in preterm neonates. The assessment is typically based upon multiple findings rather than a single isolated finding. Clinical findings (eg, respiratory support requirement) should also be considered when determining if a PDA is hemodynamically significant. Refer to UpToDate topics on PDA in preterm infants for additional details.

PDA: patent ductus arteriosus; BSA: body surface area; LPA: left pulmonary artery; Ao: aorta; LA: left atrium; LV: left ventricle; RA: right atrium; PFO: patent foramen ovale.

Graphic 141537 Version 1.0

## Our suggested approach to managing hemodynamically significant patent ductus arteriosus (PDA) in preterm neonates




This figure summarizes our suggested approach to managing preterm infants with hemodynamically significant (large) PDA on echocardiography. Echocardiographic parameters that indicate a hemodynamically significant PDA in the preterm infant include:

- PDA diameter >1.5 mm or PDA:LPA ratio ≥1
- Diastolic flow reversal in the abdominal aorta
- Unrestrictive flow across the PDA (Ao-PDA flow velocity <2 m/s or mean pressure gradient ≤8 mmHg across the PDA)
- Left atrial and/or ventricular enlargement not otherwise explained

In general, a combination of multiple findings is more suggestive than a single isolated finding. The optimal management approach for infants with hemodynamically significant PDA is uncertain, and practice varies. At the author's center, we use a step-wise management approach that begins with supportive care alone in most cases. We prefer this approach because spontaneous closure will occur in many patients. However, other centers may use different management approaches (eg, early therapy for any EPT infant with a large PDA). Refer to UpToDate topics on PDA in preterm infants for further details.

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PDA: patent ductus arteriosus; EPT: extremely preterm; GI: gastrointestinal; AKI: acute kidney injury; NEC: necrotizing enterocolitis; CHD: congenital heart disease; PMA: postmenstrual age; LPA: left pulmonary artery; Ao: aorta.

* Supportive care measures include a neutral thermal environment, optimal respiratory support, monitoring for anemia, and modest fluid restriction. For additional details, refer to separate topics on neonatal respiratory support, fluid therapy, and anemia of prematurity.

¶ Options for pharmacologic therapy include ibuprofen, acetaminophen (paracetamol), and indomethacin. The choice is center-dependent. All 3 agents appear to have similar efficacy, but the risk of adverse effects (eg, GI bleeding, AKI) is highest with indomethacin. Contraindications to ibuprofen and indomethacin include AKI/oliguria, GI bleeding, NEC (proven or suspected), thrombocytopenia and/or coagulopathy, acute infection, and ductal-dependent CHD. In most of these circumstances, if the infant requires pharmacotherapy for PDA closure, acetaminophen can be used. The exception is infants with ductal-dependent CHD who should not be given any medication to close the PDA.

Δ For patients who fail to respond to 2 courses of therapy, subsequent treatment courses are unlikely to close the PDA. If the infant remains symptomatic from the PDA (eg, ventilator-dependent), options for definitive closure include surgical ligation or transcatheter occlusion. The choice between these is dependent on the expertise available at the center. Refer to UpToDate topic on management of PDA in preterm neonates for additional details.

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Graphic 127115 Version 3.0

## Overview of mechanical ventilation in preterm neonates

### Key principles

**Minimizing VILI:** Therapeutic strategies to support gas exchange while minimizing VILI include:

- Avoidance of MV through preferential use of nCPAP when possible
- For neonates who fail nCPAP and require invasive MV, lung protective strategies include:
  - Preferential use of VTV with Tv of 4 to 6 mL/kg to minimize volutrauma
  - Use of PEEP to maintain lung recruitment and avoid atelectasis
  - Avoidance of high $FiO_2$
  - Setting targets for gas exchange that do not aim for normal levels (ie, modest permissive hypercapnia)
  - Use of HFOV or HFJV in neonates at high risk of developing VILI or as rescue therapy for neonates with refractory respiratory failure while on CMV

**Achieving gas exchange:** With CMV, the primary means of achieving ventilation ($CO_2$ clearance) and oxygenation (uptake of $O_2$) are as follows:

- Ventilation – $CO_2$ clearance is primarily determined by minute ventilation (ie, RR and Tv).
- Oxygenation – Oxygenation is primarily determined by MAP and $FiO_2$. In CMV, MAP is largely determined by PEEP.

### Indications for MV

If, despite efforts to optimize noninvasive support, the neonate develops any of the following signs of inadequate gas exchange, we typically intubate and initiate invasive MV*:

- pH <7.20, with $PaCO_2$ >65 mmHg
- Requiring $FiO_2$ >0.4 to 0.5 to achieve target $SpO_2$ goal
- Multiple apneic episodes per hour associated with desaturations and bradycardia or more than 1 episode requiring positive pressure ventilation within a few hours

### Initial mode and settings

The ventilator mode and settings must be tailored to meet the needs of the individual neonate, which may differ between patients and within the same patient over time. In our center, we typically initiate MV with CMV and reserve HFV for cases of refractory respiratory failure despite efforts to optimize CMV However, other centers may use HFV as an initial ventilation strategy in neonates at high risk for developing VILI.

**Mode:**

- We typically use a synchronized mode with both mandatory and spontaneous breaths (ie, SIMV + PS or ACV)
- We preferentially use VTV in all preterm neonates
- We use PLV if there is a technical challenge that limits the reliable delivery of measured Tv (eg, large ETT leak) or VTV is not available

**Initial settings:**

- Tv 4 to 6 mL/kg
- PEEP 5 to 6 cm $H_2O$
- Ti 0.35 to 0.4 seconds

| Monitoring |
| --- |
| Appropriate monitoring includes:<br>▪ Continuous $SpO_2$ monitoring<br>▪ Serial assessments of work of breathing<br>▪ Blood gases¶ – CBG and VBG are adequate in many cases, but placement of an arterial catheter for ABG sampling may be warranted in some cases (if the neonate is requiring blood sampling more frequently than every 6 hours or has hemodynamic instability requiring active titration of vasoactive medications)<br>▪ Ventilator monitoring, including PIP and exhaled Tv measured by the ventilator<br>▪ Chest radiographs – Chest radiographs should be obtained judiciously to inform decisions about ventilator settings and/or identify acute changes (eg, air leak, malpositioned ETT) |
| **Gas exchange targets** |
| **Oxygen target:** $SpO_2$ target 90 to 95% |
| **Carbon dioxide targets:**<br>▪ For most preterm neonates in the first few weeks of life: $pCO_2$ target is between 40 and 65 mmHg (ie, modest permissive hypercapnia)<br>▪ For older preterm infants with evolving BPD, it is reasonable to use more liberally permissive $pCO_2$ targets as long as the pH remains >7.25 |

This table summarizes the general approach to MV in preterm neonates and it reflects the practice at the author's institution. However, practice is not standardized, and other centers may use a different approach. For further details, including the evidence supporting these MV settings and targets, refer to UpToDate content on MV in neonates.

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VILI: ventilator-induced lung injury; MV: mechanical ventilation; nCPAP: nasal continuous positive airway pressure; VTV: volume-targeted ventilation; Tv: tidal volume; PEEP: positive end-expiratory pressure; $FiO_2$: fraction of inspired oxygen; HFOV: high-frequency oscillatory ventilation; HFJV: high-frequency jet ventilation; CMV: conventional mechanical ventilation; RR: respiratory rate; MAP: mean airway pressure; $PaCO_2$: partial pressure of arterial carbon dioxide; $SpO_2$: peripheral oxygen saturation; HFV: high-frequency ventilation; SIMV + PS: synchronized intermittent mandatory ventilation plus pressure support; ACV: assist control ventilation; PLV: pressure-limited ventilation; ETT: endotracheal tube; Ti: inspiratory time; CBG: capillary blood gas; VBG: venous blood gas; ABG: arterial blood gas; PIP: peak inspiratory pressure; $pCO_2$: partial pressure of carbon dioxide.

* We rely mostly on these objective measures of gas exchange to define CPAP failure. However, the thresholds are not hard set, and these parameters should be interpreted in conjunction with other clinical findings. For neonates with signs of labored breathing, hemodynamic instability, or persistent metabolic acidosis, we generally use a lower threshold for transitioning to invasive MV.

¶ Transcutaneous carbon dioxide monitoring (TCOM) may be used in select circumstances (eg, neonates with severely compromised ventilation or when dynamic changes in $pCO_2$ levels are anticipated, particularly when transitioning to, or titrating, HFOV). Other centers use TCOM routinely.

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Graphic 131155 Version 1.0

## Our suggested thresholds for RBC transfusion in neonates

**Clinical instability is defined by the presence of *at least one* of the following:**

- Invasive mechanical ventilation
- Circulatory failure requiring inotropic/vasopressor support
- CPAP or other non-invasive positive pressure ventilation support with fraction of inspired oxygen >0.4
- Acute sepsis or necrotizing enterocolitis with hemodynamic instability requiring pressor support
- >6 documented apneas requiring moderate intervention per 24 hours
- Undergoing major surgery (within or up to 48 hours after surgery)

| Postnatal age | Gestational age | Hemoglobin (hematocrit) thresholds for transfusion | |
|---|---|---|---|
| | | Neonates *with* clinical instability | Neonates *without* clinical instability |
| 0-7 days | Any | 11 g/dL (32%) | 10 g/dL (29%) |
| 8-14 days | <35 weeks | 10 g/dL (29%) | 8 g/dL (24%) |
| | ≥35 weeks | 7 g/dL (21%)* | 7 g/dL (21%)* |
| ≥15 days | <35 weeks | 8 g/dL (24%) | 7 g/dL (21%)¶ |
| | ≥35 weeks | 7 g/dL (21%)* | 7 g/dL (21%)* |

This table summarizes our suggested hemoglobin and hematocrit thresholds for considering a RBC transfusion in neonates based upon postnatal age and clinical status. These thresholds are appropriate for term and preterm neonates cared for in the NICU. These thresholds do **not** apply to neonates with cyanotic congenital heart disease, severe alloimmune hemolytic disease, acute severe or ongoing blood loss, severe PPHN, or those on ECMO. For additional details, refer to separate UpToDate content on RBC transfusions in neonates.

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CPAP: continuous positive airway pressure; RBC: red blood cell; NICU: neonatal intensive care unit; ECMO: extracorporeal membrane oxygenation; AOP: anemia of prematurity; PPHN: persistent pulmonary hypertension of the newborn; Hgb: hemoglobin; HCT: hematocrit; SGA: small for gestational age.

* This threshold is appropriate for most neonates in these categories, provided they are not SGA (in which case the preterm thresholds should be used). The Hgb <7 g/dL (HCT <21%) threshold applies to clinically stable neonates and to critically ill neonates who are hemodynamically stabilized (ie, not hypotensive and not requiring escalating inotropic/vasopressor support). A higher threshold may be warranted in neonates with severe or progressive shock and/or severe hypoxemia. In addition, separate thresholds are used for neonates with any of the conditions listed above in the legend.

¶ For preterm neonates with AOP who otherwise appear well and are asymptomatic but who continue to have significant anemia (ie, Hgb <7 g/dL or HCT <21%) at four to six weeks after birth, measuring the

reticulocyte count can help guide decisions regarding RBC transfusion. If the absolute reticulocyte is ≥100,000/microL (≥2%), RBC transfusion may not be necessary.

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Graphic 140490 Version 3.0

## Contributor Disclosures

**Joseph B Philips III, MD, FAAP** No relevant financial relationship(s) with ineligible companies to disclose. **Joseph A Garcia-Prats, MD** No relevant financial relationship(s) with ineligible companies to disclose. **David R Fulton, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Niloufar Tehrani, MD** No relevant financial relationship(s) with ineligible companies to disclose.

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